Smart Parking

By Nichita Marcus

Cuprins

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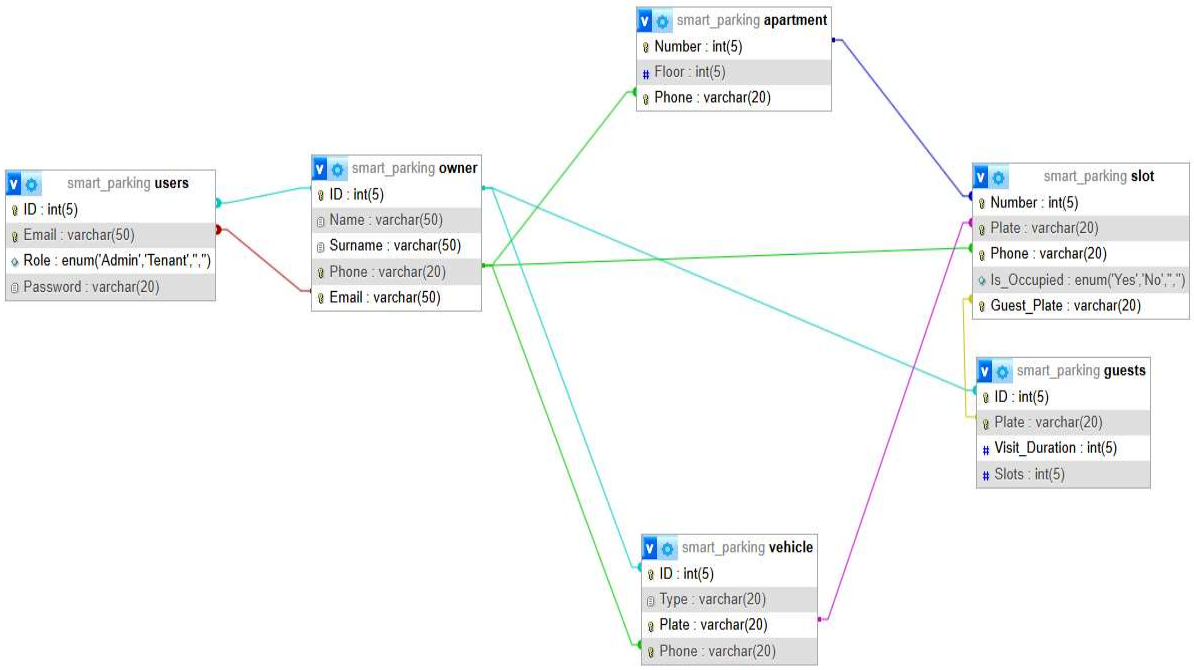
🏢 Project Description

This project is designed for managing a **residential parking lot system** tailored to apartment complexes. It focuses on organizing and streamlining parking for both **residents (tenants)** and their **guests**, using a relational database that ensures secure access and efficient slot allocation.

👨‍💼 User Roles and Access

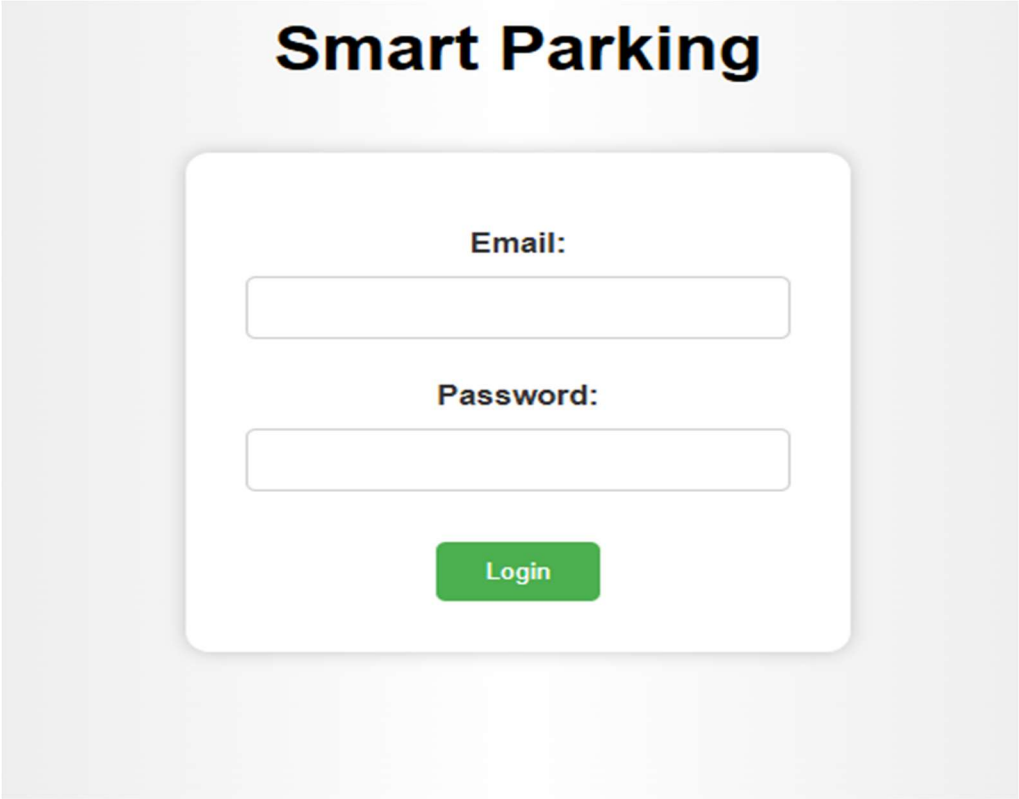
- The **Administrator (Landlord)** is the only user with full access to the database. They can view and manage all entries.
- **Tenants (Apartment Residents)** must log in using their credentials and provide required data such as personal, apartment, and vehicle information. They can only view and manage their own data.

🧠 Database Diagram



🌐 Before and After of Views and Website Appearance

Login menu:



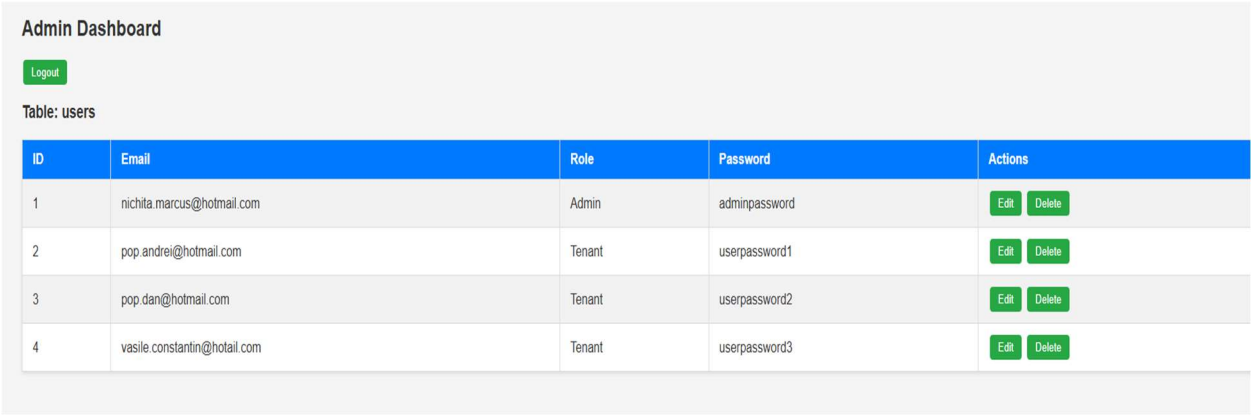
Login code:

```
$admin_email = "nichita.marcus@hotmail.com";
$admin_pass = "adminpassword";

$user_email1 = "pop.andrei@hotmail.com";
$user_pass1 = "userpassword1";
$user_email2 = "pop.dan@hotmail.com";
$user_pass2 = "userpassword2";
$user_email3 = "vasile.constantin@hotmail.com";
$user_pass3 = "userpassword3";

if ($email === $admin_email && $password === $admin_pass) {
    $_SESSION['email'] = $email;
    $_SESSION['role'] = 'Admin';
    header("Location: admin_dashboard.php");
    exit;
} elseif ($email === $user_email1 && $password === $user_pass1) {
    $_SESSION['email'] = $email;
    $_SESSION['role'] = 'Tenant';
    header("Location: user_dashboard.php");
    exit;
} elseif ($email === $user_email2 && $password === $user_pass2) {
    $_SESSION['email'] = $email;
    $_SESSION['role'] = 'Tenant';
    header("Location: user_dashboard.php");
    exit;
}elseif ($email === $user_email3 && $password === $user_pass3) {
    $_SESSION['email'] = $email;
    $_SESSION['role'] = 'Tenant';
    header("Location: user_dashboard.php");
```

Admin Dashboard where admin can edit or delete data:



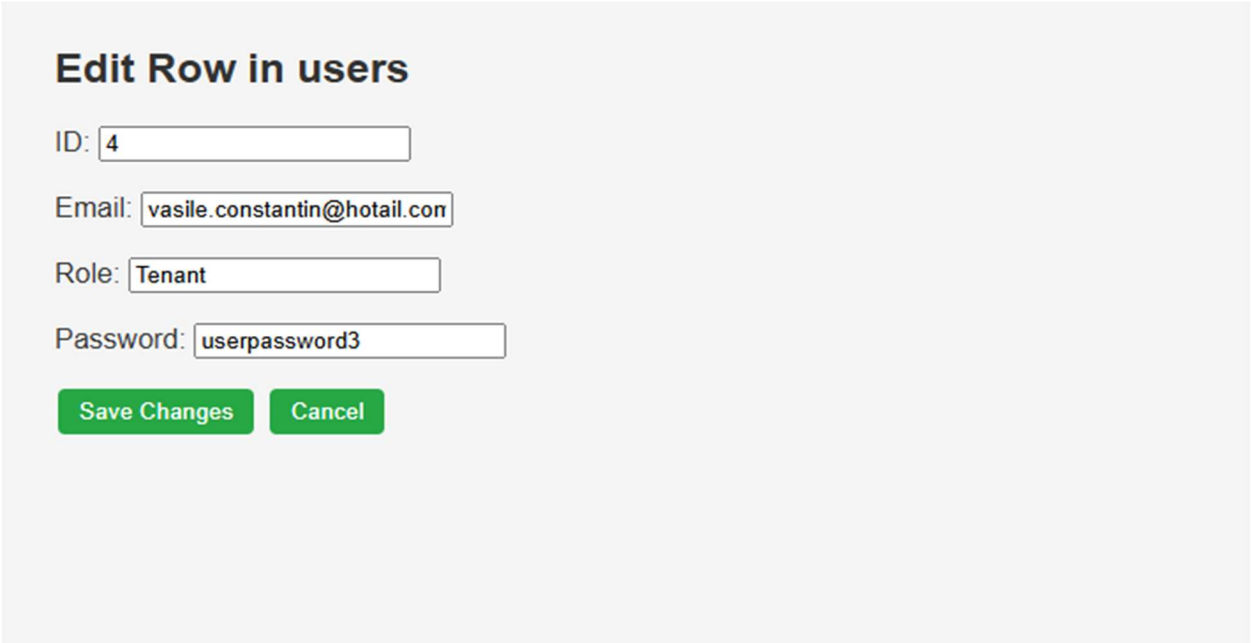
Admin Dashboard

Logout

Table: users

ID	Email	Role	Password	Actions
1	nichita.marcus@hotmail.com	Admin	adminpassword	Edit Delete
2	pop.andrei@hotmail.com	Tenant	userpassword1	Edit Delete
3	pop.dan@hotmail.com	Tenant	userpassword2	Edit Delete
4	vasile.constantin@hotail.com	Tenant	userpassword3	Edit Delete

Admin edit menu:



Edit Row in users

ID: 4

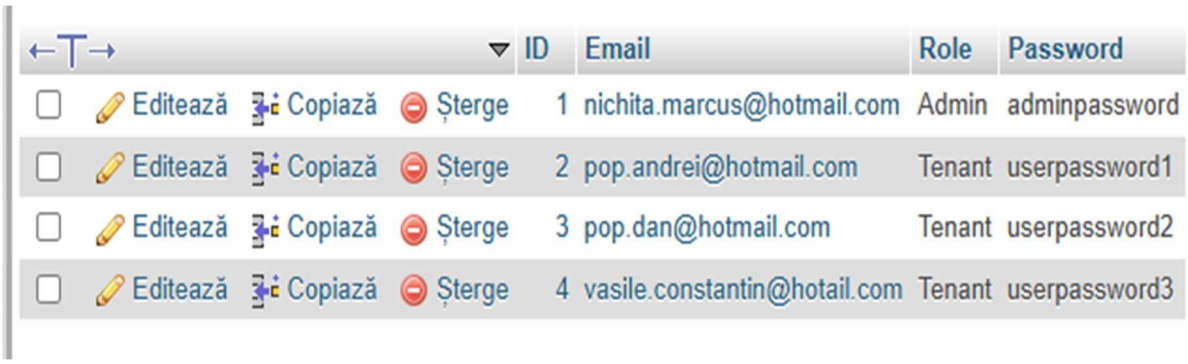
Email: vasile.constantin@hotail.con

Role: Tenant

Password: userpassword3

Save Changes Cancel

User table before editing: **vasile.constantin@hotail.com**



←T→			ID	Email	Role	Password
Editează	Copiază	Șterge	1	nichita.marcus@hotmail.com	Admin	adminpassword
Editează	Copiază	Șterge	2	pop.andrei@hotmail.com	Tenant	userpassword1
Editează	Copiază	Șterge	3	pop.dan@hotmail.com	Tenant	userpassword2
Editează	Copiază	Șterge	4	vasile.constantin@hotail.com	Tenant	userpassword3

User table after editing: **vasile.constantin@hotmail.com**

←T→			ID	Email	Role	Password
Editează	Copiază	Șterge	1	nichita.marcus@hotmail.com	Admin	adminpassword
Editează	Copiază	Șterge	2	pop.andrei@hotmail.com	Tenant	userpassword1
Editează	Copiază	Șterge	3	pop.dan@hotmail.com	Tenant	userpassword2
Editează	Copiază	Șterge	4	vasile.constantin@hotmail.com	Tenant	userpassword3

User table after delete:

←T→			ID	Email	Role	Password
Editează	Copiază	Șterge	1	nichita.marcus@hotmail.com	Admin	adminpassword
Editează	Copiază	Șterge	2	pop.andrei@hotmail.com	Tenant	userpassword1
Editează	Copiază	Șterge	3	pop.dan@hotmail.com	Tenant	userpassword2

User Dashboard before adding guest:

User Dashboard

owner

ID	Name	Surname	Phone	Email
3	Pop	Dan	0764230336	pop.dan@hotmail.com

vehicle

ID	Type	Plate	Phone
3	Motor Bike	CJ03DAN	0764230336

apartment

Number	Floor	Phone
11	1	0764230336

slot

Number	Plate	Phone	Is_Occupied	Guest_Plate
11	CJ03DAN	0764230336	Yes	

guests

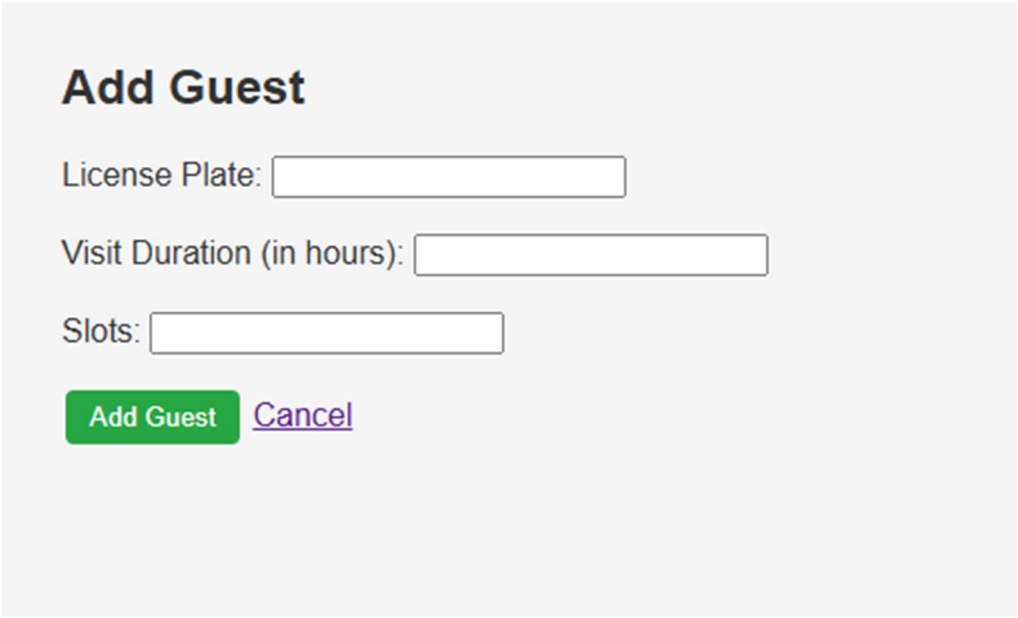
Add Guest

Guest table before adding guest:

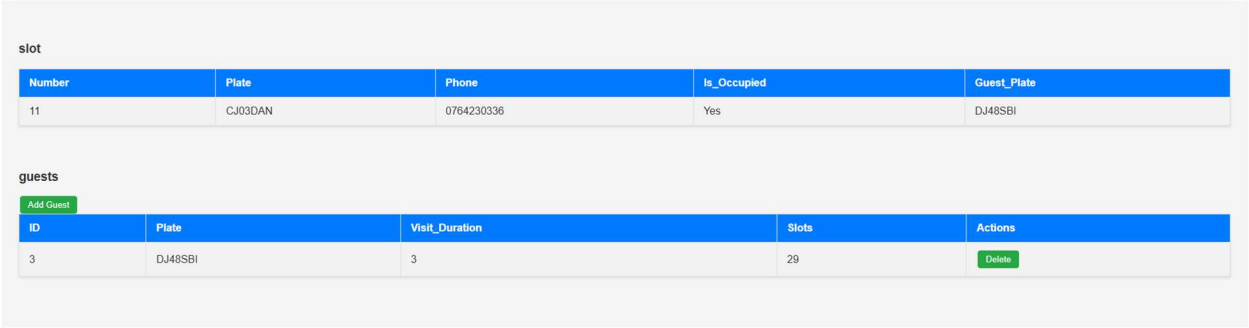
Operații rezultate interogări

Creează vizualizarea

User menu for adding guest:



User Dashboard after adding guest:



slot

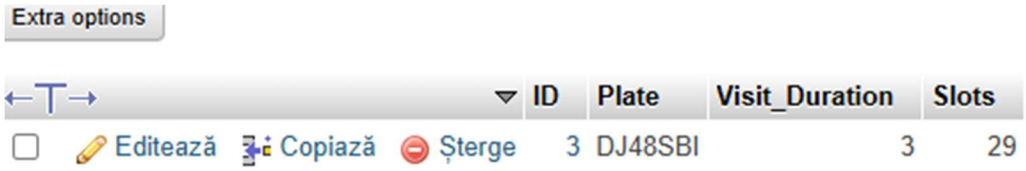
Number	Plate	Phone	Is_Occupied	Guest_Plate
11	CJ03DAN	0764230336	Yes	DJ48SBI

guests

Add Guest

ID	Plate	Visit_Duration	Slots	Actions
3	DJ48SBI	3	29	Delete

Guest table after adding guest:



🗂️ Database Tables and Purpose

1. **users**
    - Stores system login data.
    - Fields: ID, Email, Password, Role (Admin, Tenant)
    - Role-based access ensures that tenants and the administrator see only relevant content.
2. **owner**
    - Contains the personal details of each apartment owner.
    - Fields: ID, Name, Surname, Phone, Email
    - The Phone number acts as the primary link to related data.
3. **apartment**
    - Stores apartment-specific information.
    - Fields: Number, Floor, Phone
    - Each apartment is linked to its owner via the phone number.
    - Each apartment has one dedicated parking slot, matching the apartment number.
4. **vehicle**
    - Contains data about each tenant's vehicle.
    - Fields: ID, Type, Plate, Phone
    - Vehicles are associated with owners using their phone number.
5. **slot**
    - Manages parking slot assignments.

- Fields: Number, Plate, Phone, Is_Occupied (YES / NO), Guest_Plate
- Each vehicle is assigned a slot based on its apartment.
- Also tracks guest usage of slots via the Guest_Plate field.

6. **guests**

- Tracks temporary guest vehicle access.
- Fields: ID (owner ID), Plate, Visit_Duration, Slots
- Guest entries are linked to the apartment they are visiting via the owner's ID.
- The database ensures a total of 30 guest slots across the complex.
- Tenants can only delete guest entries associated with their ID.

🔁 Key Design Considerations

- The Phone number in the owner table is the critical field, acting as the central link across multiple tables (vehicle, apartment, slot).
- Updating the phone number requires contacting the Administrator, as it affects all linked records.
- Tenants can add and delete their own guest entries, but only the Administrator can manage global records.

📚 Bibliography

- StackOverflow